



Lab 01: C Fundamentals

CSE 4108

Structured Programming I Lab

Lab Tasks

1. Value Added Tax:

Bvaly is a multinational business venture that focuses on e-commerce. To cover the maintenance cost, they charge 5% VAT for any item purchased. For example: If you buy a product worth \$100.00, you have to pay \$105.00 (including VAT). They have hired you to automate the process of calculating the amount a customer needs to pay. Write a C program that asks the user to enter the dollar-and-cents amount, then displays the amount with 5% VAT added.

Sample run:

```
Enter an amount: 100.00  
With VAT added: 105.00
```

Note that, underlined value denotes user input.

2. Polynomial Evaluation:

Write a C program that asks the user to enter the value of x and outputs the value of the following polynomial:

$$3x^5 - 2x^4 + 5x^3 + x^2 - 7x + 6$$

Sample run:

Enter value of x: 2

Value of the polynomial for x=2: 100.00

*Hint: C doesn't have an exponentiation operator, so you'll need to multiply x by itself repeatedly in order to compute the power of x. (For example: $x*x*x$ is x cubed)*

3. Horner's Rule:

Modify the program of the last task so that the polynomial is evaluated using the following formula:

$$((((3x - 2)x + 5)x + 1)x - 7)x + 6$$

Note that the modified program performs fewer multiplications. This technique for evaluating polynomials is known as *Horner's Rule*.

4. Distance:

Two common distance metrics that are widely used when calculating distance between two points are the **Manhattan distance**, otherwise known as the L^1 distance, and **Euclidean distance**, otherwise known as the L^2 distance. Mathematically, these are given as follows:

$$L^1 = |x_2 - x_1| + |y_2 - y_1|$$

$$L^2 = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

You are given two points, (x_1, y_1) and (x_2, y_2) , in the XY plane. Show the Manhattan distance and Euclidean distance between the given points.

Sample run:

```
Enter x1 and y1: -1 3
Enter x2 and y2: 3 2
Manhattan distance: 5.000000
Euclidean distance: 4.123106
```

Hint: Some of the necessary functions here will require you to include the `math.h` header file. You can find the absolute value using the `fabs(double)` [e.g., `fabs(-5.5)` will return 5.5] function and similarly square root of a value using the `sqrt(double)` [e.g., `sqrt(2.0f)` will return 1.4142136 (or a value close to this one)] function.

5. Leave Me A Loan:

ABC Bank Ltd. provides loans to its customers. The customers return the amount by a fixed monthly payment. On each month however, the remaining balance is incremented by the

interest rate assigned for each customer. The total balance at the end of each month can be calculated using the following formula:

$$\text{currBalance} = \text{prevBalance} - \text{pay} + (\text{prevBalance} \times \text{rate})$$

Here,

currBalance = Updated Balance

prevBalance = Previous Balance

pay = Monthly Payment

rate = Monthly Interest Rate

Your task is to write a program that calculates the remaining payable amount on a loan after the first, second, and third monthly payment.

Your program will take the loan amount, yearly interest rate (in percent), fixed monthly payment as input. You need to display each balance with two digits after the decimal point.

Sample Run:

Enter amount of loan: 20000.00

Enter interest rate: 6.0

Enter monthly payment: 386.66

Balance remaining after first payment: \$19713.34

Balance remaining after second payment: \$19425.25

Balance remaining after third payment: \$19135.71

Hint: Note the interest rate that is given as input is the yearly interest rate in percent. To find the monthly interest rate, convert the interest rate entered by the user to a fractional value (e.g., 10 percent becomes 0.1) and divide it by **12**.